



Radiology of The Urogenital System

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Learning objectives

- Medical students will be able to identify the types of radiological examinations used to diagnose urinary tract disease.
- Medical students will learn how to evaluate BNO, IVP, and urethrocytography images.

Radiological Examination of the Urinary Tract

- BNO
- BNO-IVP
- Urethrocystography
- Retrograde pyelography
- Micturation cystouretrografi
- Ultrasound
- MSCT scan urography
- MRI urography
- Nuclear

Anatomy

Urinary Tract

- Kidney
- Ureter
- Bladder/vesika urinarius
- Urethra

Female genitalia

- Vagina
- Uterus
- Tuba Fallopi
- Ovarium

Male genitalia

- Testis dan epididymis
- Vas deferens
- Prostat
- Penis

Indication

- Hematuria
- Dysuria
- Urinary retention
- Ureteric colic Abdominal/flank
- Pelvic mass accompanied by pain and/or hematuria
- Recurrent infections
- Congenital abnormalities (renal agenesis, horse shoe kidney, duplicating collecting system)
- Post trauma

Plain abdomen (BNO)

- Plain abdominal radiograph (BNO) (Kidney Ureter Bladder)
- This Plain abdominal photo image is used to view the urinary tract, from the kidneys to the bladder.
- Indication: To check for radiopaque stones in the urinary tract.

When receiving the results of the radiological examination

- Patient Identity Check Name, Gender, Age, and Medical Record Number
- Photo ID Check Registration Number and Issue Date
- Marker Check (R/L or D/S)

BNO

- Check the condition of the image.
 - The BNO image is sufficient for T12-symphysis pubis.
 - Note the fecal material (related to patient preparation).

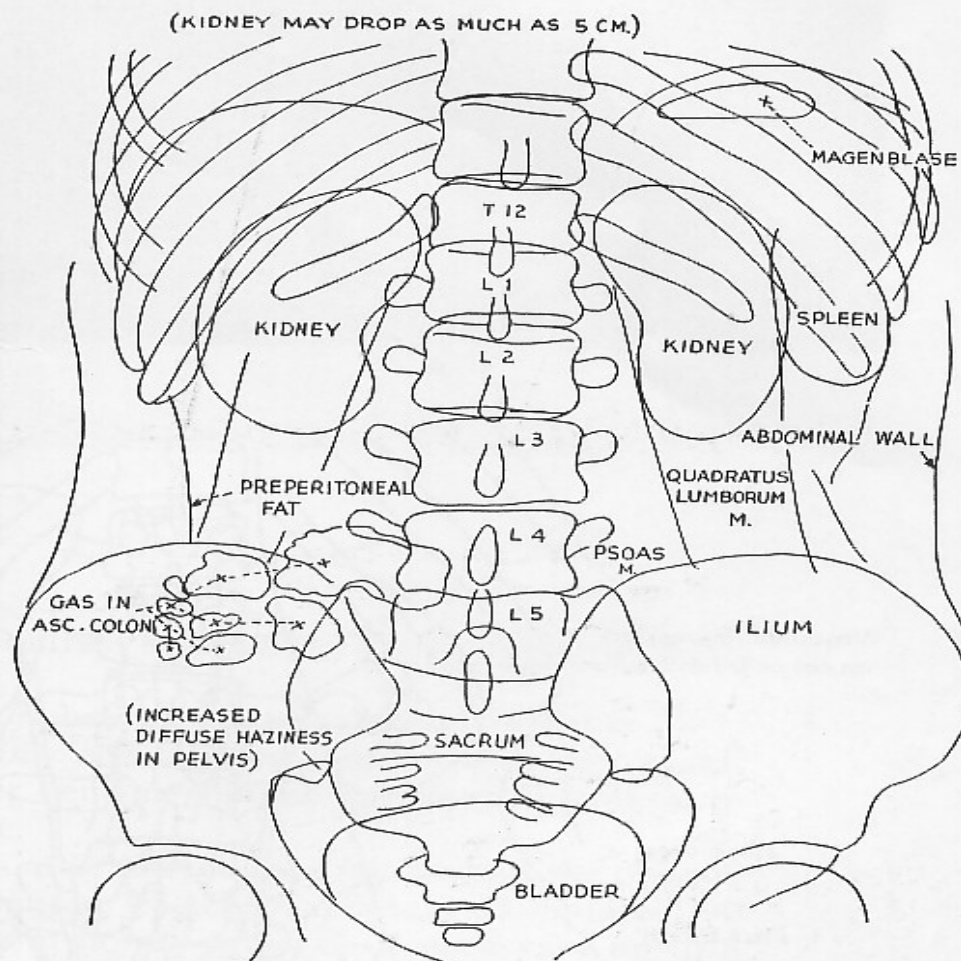
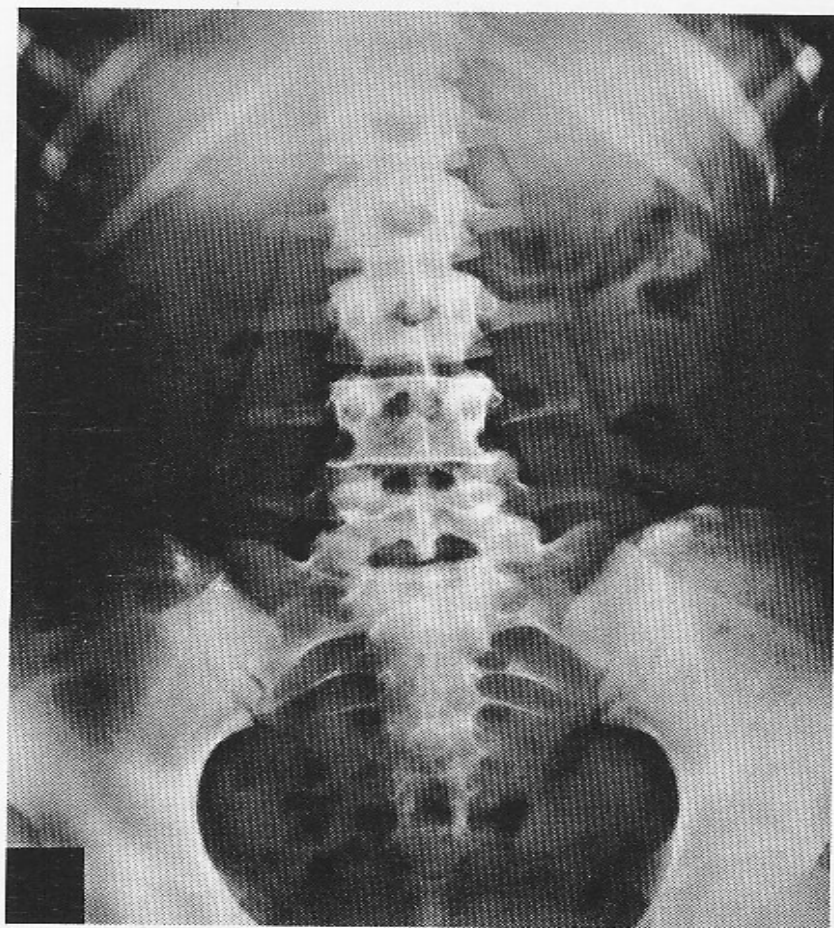
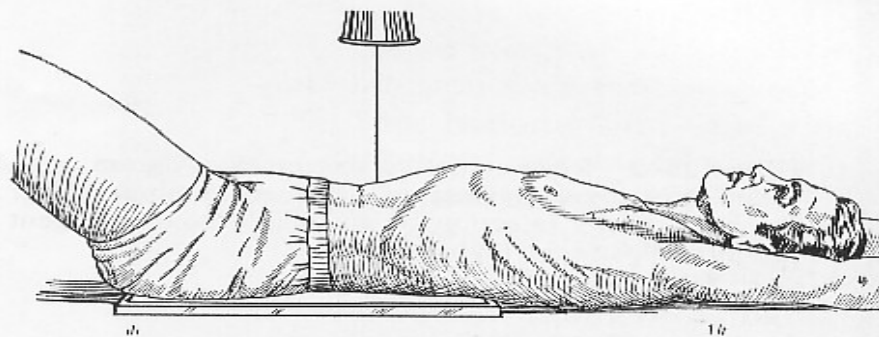


Figure 25-1 Antero-posterior view of abdomen (KUB film).

Evaluate BNO photos

- Identify the kidney contour
- Identify the presence of radiopaque shadows in the urinary tract: kidneys, ureteral tract, and bladder
- Other : Presence or absence of dilated intestinal loops Right and left psoas lines Right and left preperitoneal fat lines Bones

Normal BNO photo





Nephrolith



Nephrolith



Ureterolith



Vesicolith



Multiple vesicolith



Staghorn renal
stone



Vesicolith

BNO-IVP

- IVP (Intravenous Pyelography)
- Examination using IV contrast.
- Purpose: Assess kidney function.
- Can detect semi-opaque or non-opaque stones that cannot be seen on a fluoroscopic X-ray.
- A fluoroscopic X-ray is always performed before an IVP.
- Allergy testing is performed.
- Serial X-rays with fluoroscopy. 5 minutes, 15 minutes, 30 minutes, and 60 minutes.

BNO IVP Examination

- Laboratory Tests Performed
 - Urea (N= 4-45 mg/dl)
 - Creatinine (N= 0.6-1.1 mg/dl)
- Skin Test -- > Allergies
- Premedication (anti-allergic)

BNO-IVP (*Blass Nier Oversich ± Intra Venous Pyelogram*)

Persiapan Pasien

- Pasien makan bubur kecap saja sejak 2 hari (48 jam) sebelum pemeriksaan BNO-IVP dilakukan.
- Jam 20.00 pasien minum garam inggris (magnesium sulfat), dicampur 1 gelas air matang sbg laxative, disertai minum air putih 1-2 gelas, terus puasa.
- Selama puasa pasien dianjurkan untuk tidak merokok dan banyak bicara guna meminimalisir udara dalam usus.

CONTRAINDICATIONS

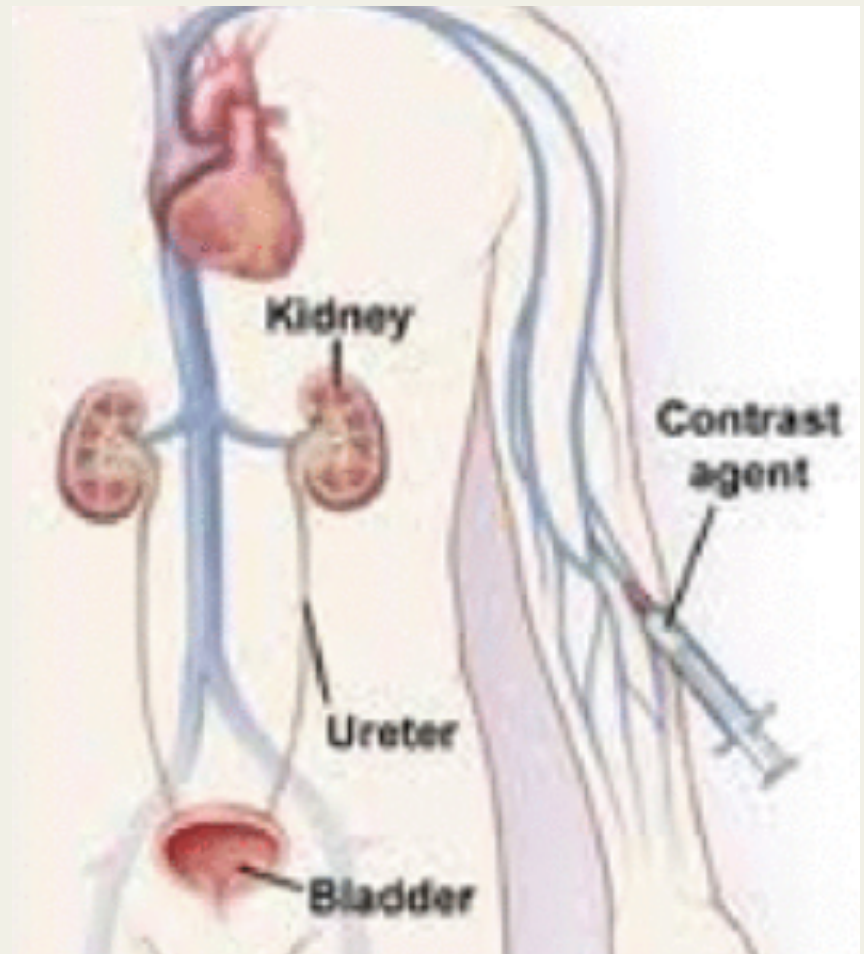
- Allergy to contrast media
- Severe heart disease Uncontrolled diabetes mellitus
- Abnormal urea and creatinine laboratory

Contrast Pathway

Contrast enters cubital vein → heart → abdominal aorta → a. renalis → a. lobularis → a. interlobularis → a. arcuate → afferent → glomerulus

Contrast appears thin

- Impaired kidney function
- Excretion impaired to obstruction
- Poor dehydration preparation



BNO-IVP Evaluation

- Evaluate the shape, size, and location of the kidneys.
- Evaluate the excretory and secretory functions of both kidneys
- Evaluate the pelviocalyceal system of both kidneys.
- Evaluate the shape, size, and signs of obstruction.
- Evaluate the tract of both ureters for dilation and stenosis (cut-off). Note the urinary bladder.

IVP Photo Series 5 minutes



5 minute photo

- Assessing contrast entry into the pelvicalyceal system (PCS) of both kidneys
- In this photo the right nephrogram is good, while the left is delayed.

IVP Photo Series 15 minutes



15 minutes Photo

- The right and left pelvic systems are much more clearly defined.
- The proximal ureter can be evaluated.
- In this photo the right pelvicocalices system is good, the left is visualized as flattening

IVP Photo Series 30 minutes



30 Minutes Photo

- No filling defects or additional defects were seen in the urinary bladder.
- The bladder mucosa was smooth, with no masses or stones visible.
- A distal blockage in the left ureter is visible, with an oval stone coated with contrast and slightly enlarged in caliber. The left pelvic caliceal system is slightly dilated.

IVP Photo Series 45 - 60 minutes

45- 60 minutes Photo



- Contrast fills the VU further.
- If there is a delay in filling the PCS, it will be re-evaluated until the 60th minute.

IVP Photo Rest contrast



Post miksi

Evaluate rest kontrast

In this photo, the remaining contrast is minimal in the VU, there is still a lot in the left ren



Grade I



Grade II

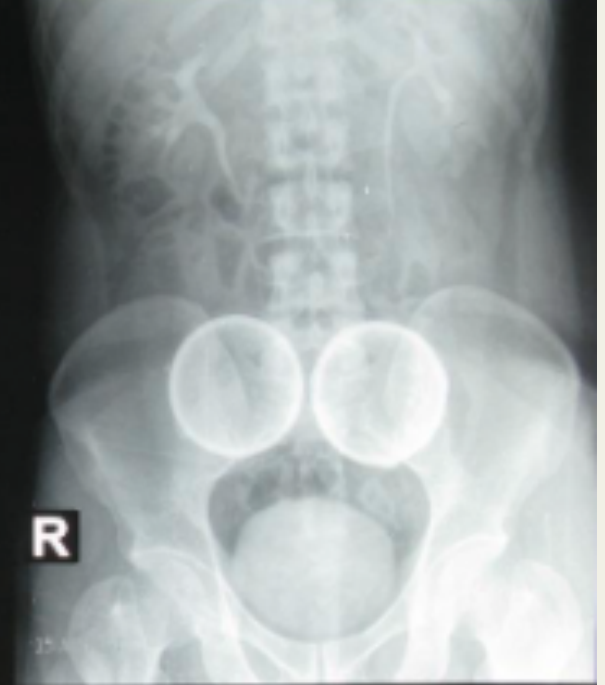


Grade III

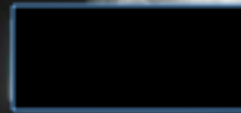


Grade IV

Normal BNO-IVP

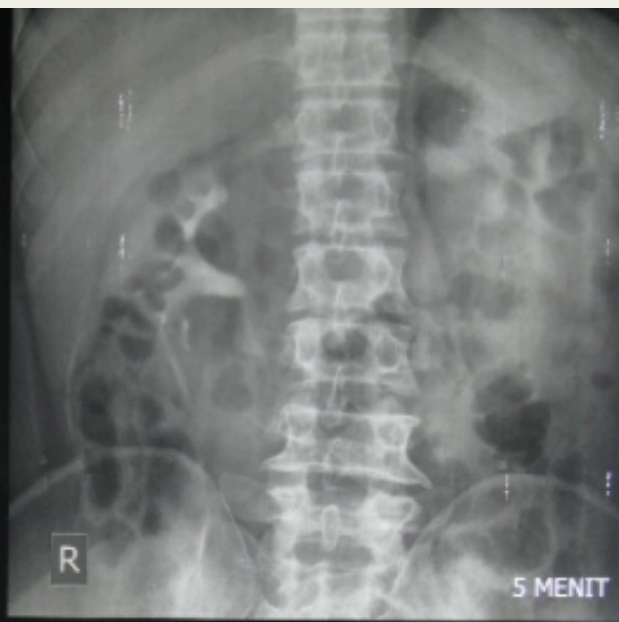








30/05/13 10:30:00 45.9 %



30/05/13 11:08:59 54.0 %



16 MENIT



30 MENIT



17/03/2016 13:02:56
Abdomen AP

EI: 347
51.6 %



17/03/2016 13:17:56
Abdomen AP

EI: 316
51.9 %



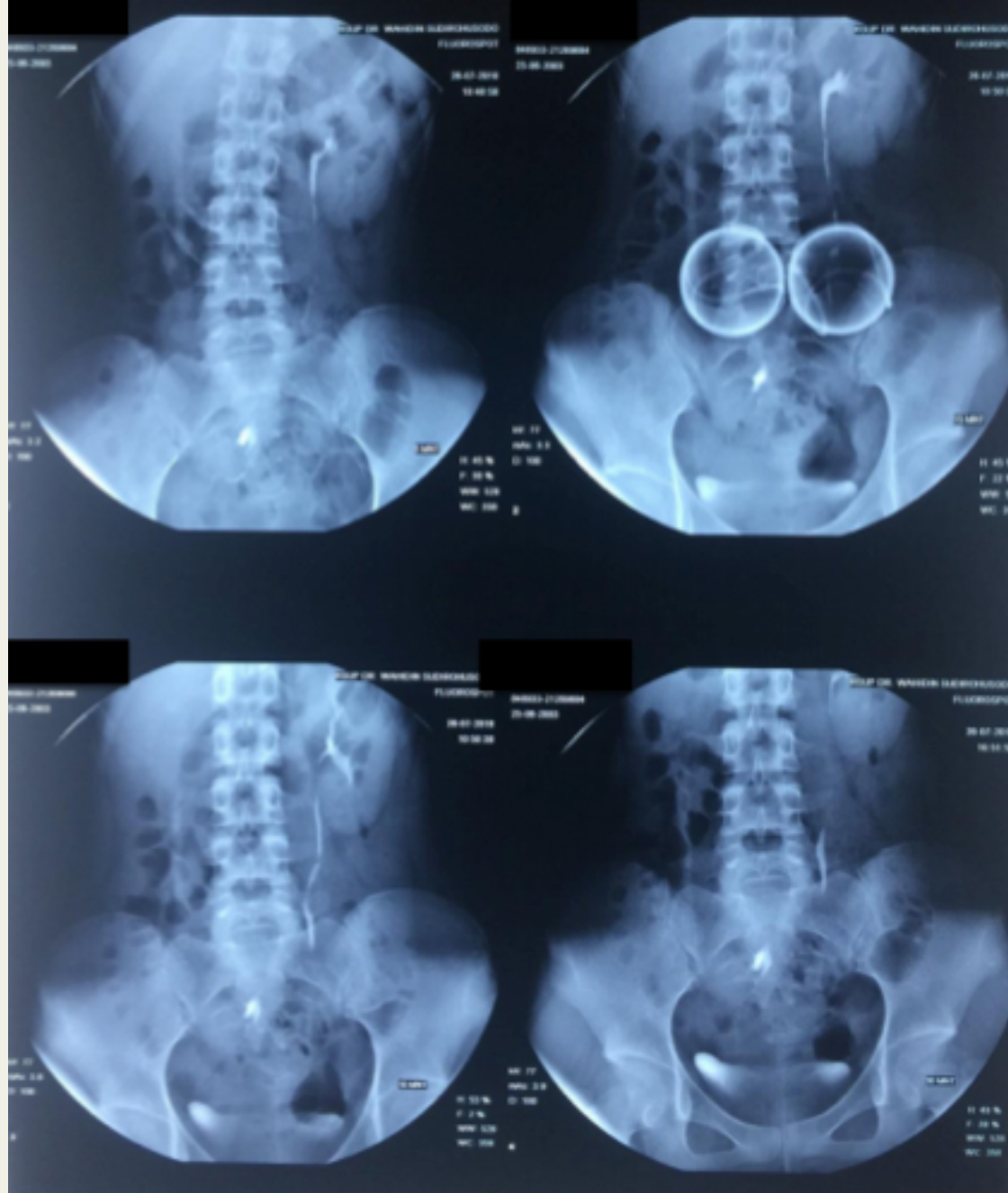




Hypertrophic prostate



BNO



IVP



Duplicating collecting system

Uretrocystography

The examination uses contrast injected through the external urethral orifice.

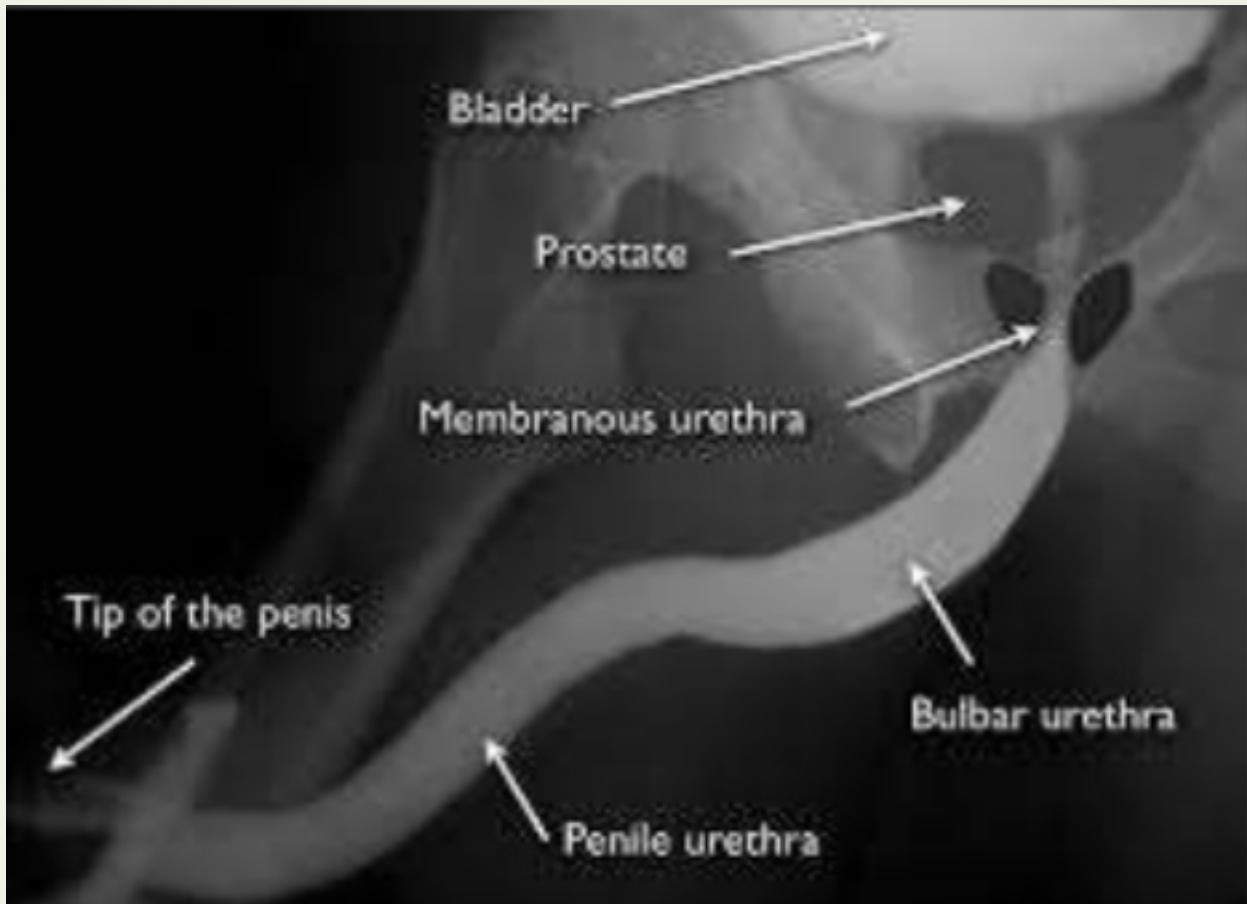
Indications

- Cystitis (inflammation of the bladder)
- Obstruction (blockage) → stricture Bladder Calculi
- Vesicolithiasis (the presence of stones in the urinary bladder)
- Fistula (an abnormal opening between two internal organs or between an organ and the body's surface)
- Urethral or bladder rupture
- Hypertrophic prostat
- Bladder carcinoma

- **Retrograde Urethrocytography**
Contrast agent is injected retrogradely through the external urethral orifice.

- **Bipolar Urethrocytography**
Contrast agent is injected retrogradely from the distal to the proximal urethra, and is also injected through a cystostomy catheter.

Anatomi Uretrocystography







12/03/2016 23:04:53
Abdomen AP

EI: 416
51.9 %



R LAT

12/03/2016 23:16:02
Abdomen AP

EI: 469
56.1 %



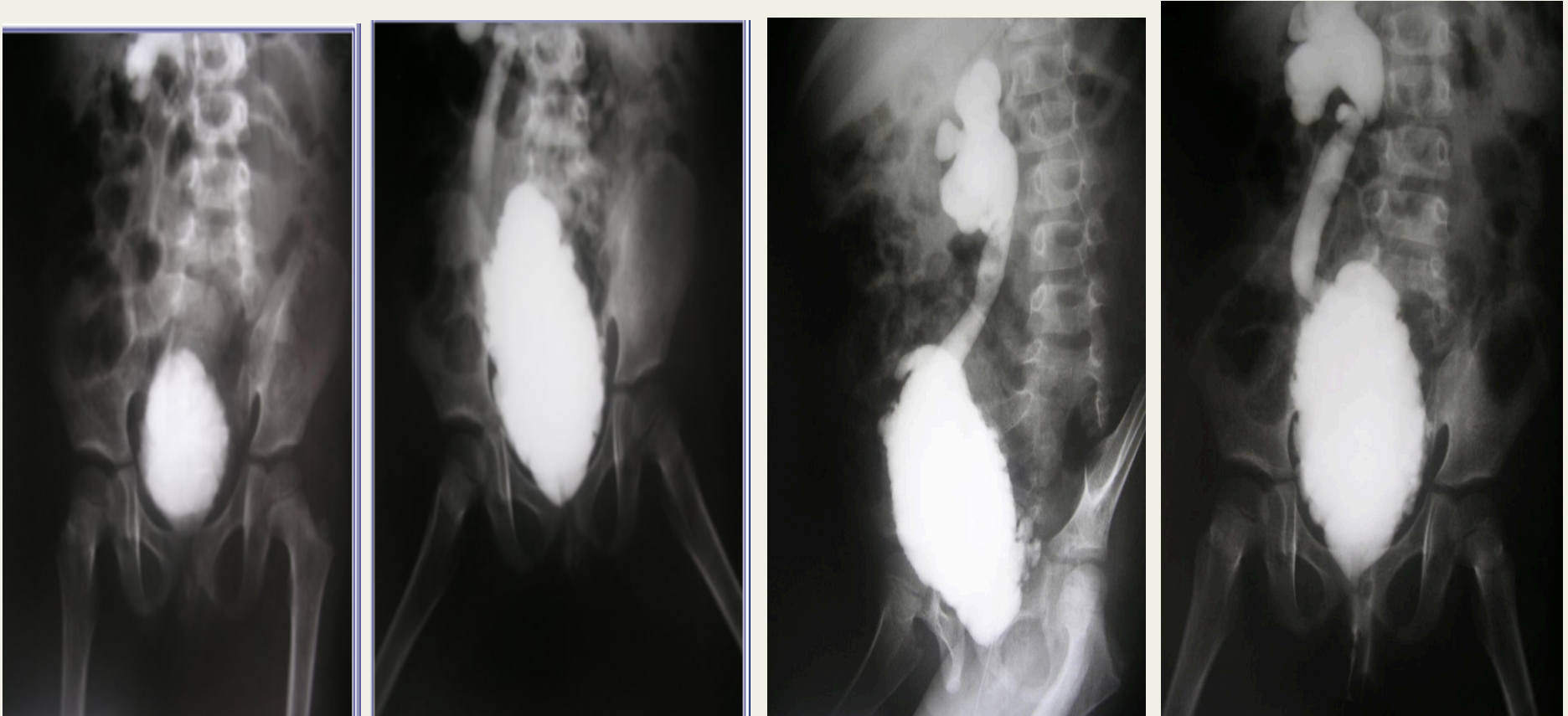
R



R

Voiding cystourethrography

- Voiding cystourethrography (VCUG), also known as a micturating cystourethrography (MCU), is a fluoroscopic study of the lower urinary tract in which contrast is introduced into the bladder via a catheter.
- The purpose of the examination is to assess the bladder, urethra, postoperative anatomy and micturition in order to determine the presence or absence of bladder and urethral abnormalities, including vesicoureteric reflux (VUR)



Abnormal cone shaped bladder dengan severe hidronefrosis disertai multipel diverticulosis
Refluks kontras ke ureter kanan sampai PCS ginjal kanan dengan gambaran severe hidronefrosis

Retrograde Pyelography (RPG)

- Ultrasound and CT cannot be performed
- Failure to visualize one or both kidneys after IVU
- Visualization of detailed calyces that are not visible on IVU
- Visualization of obstruction after IVU.
- Calcification in the ureter that is not visible on IVU

Contrast is administered via a catheter inserted through a cystoscope.



Ultrasonography

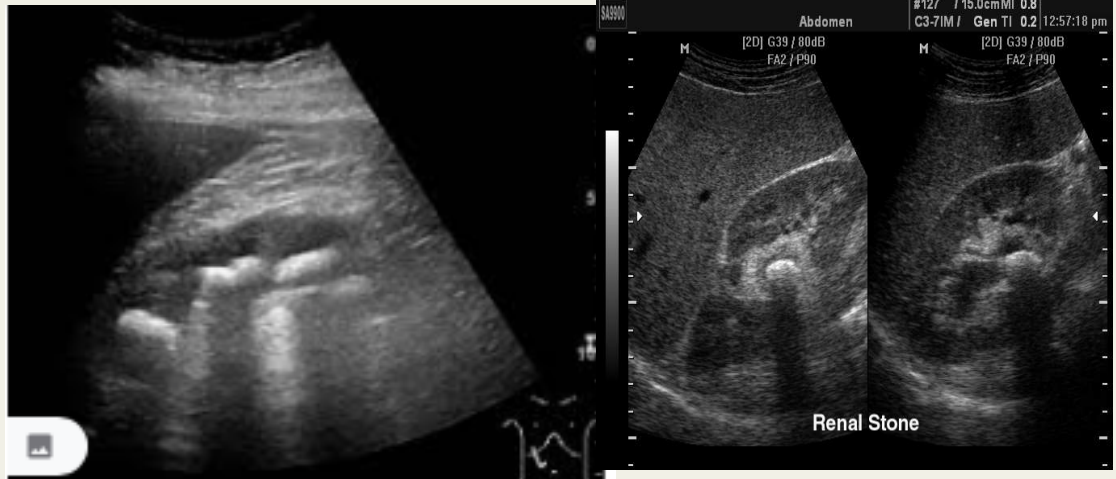
- Easy to perform
- The patient is instructed to hold their urine. Uroliths can be easily seen.
- Ureteric stones cannot be seen directly.
- The visible result (blockage) is dilation of the ureter (hydroureter) and dilation of the pelvicalyceal system (hydronephrosis).
- Kidney tumors, kidney cysts, urinary bladder tumors (tumors around the urinary bladder), and prostatic hypertrophy.



Hydronephrosis



Vesicolith

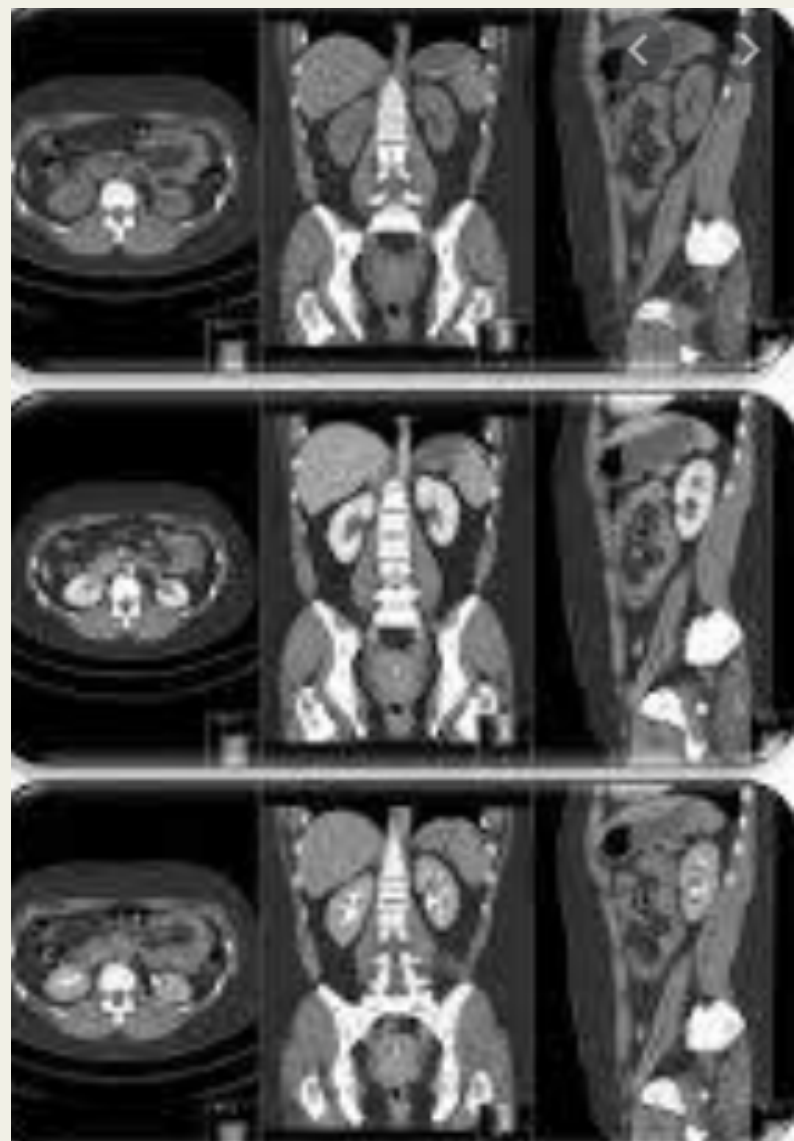


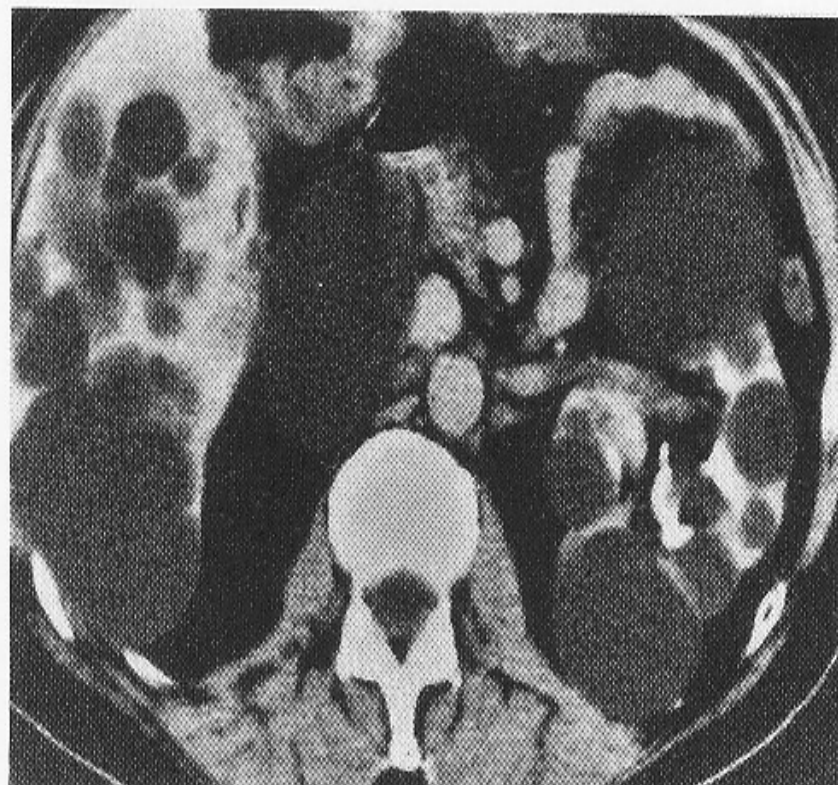
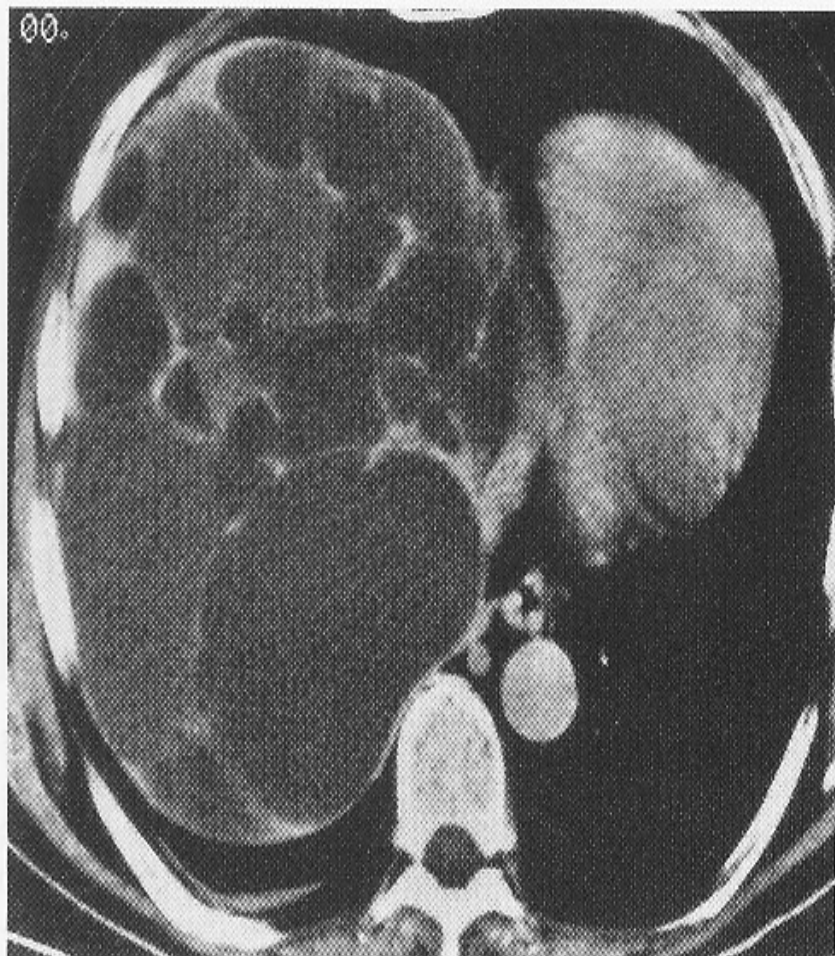
Nephrolith

CT scan Urografi

- **CT urography (CTU or CT IVU)**, also known as **CT intravenous pyelography (CT IVP)**, has now largely replaced BNO IVP in imaging the genitourinary tract. It gives both anatomical and functional information, albeit with a relatively higher dose of radiation.
- **Indications**
 - Hematuria
 - Urolithial mass
 - Obstructive uropathy (Hydronephrosis Hydroureter)
 - Urothelia cancer
 - Stricture
 - Congenital coelctyng system
- CTU has high sensitivity (96-100%) and high specificity (94-100%) of identifying ureteric and bladder calculi

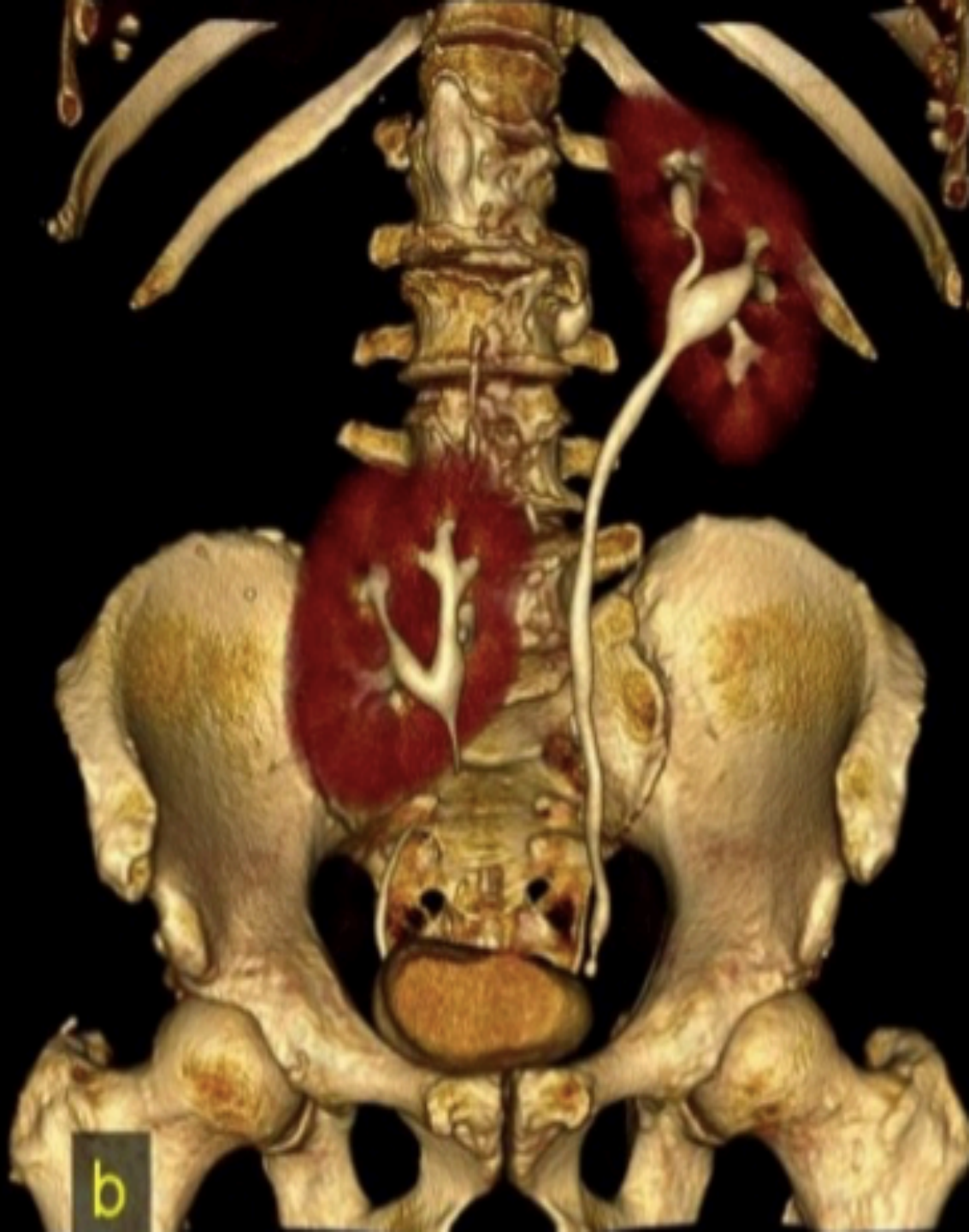
- **Purpose** The aim is to illustrate the collecting systems, ureters and bladder with intravenous contrast
- Visualization of other structures in the abdomen is also better with CTU than with traditional IVU.
- Upper tract tumors, strictures and to a degree the function of the kidney can be assessed.
- Can view the renal parenchyma
- CT scan performed when an IVP ultrasound examination reveals an unclear abnormality, such as a kidney tumor, a kidney tumor, or a bladder tumor.
- CT scan Can assess and quantify the tumor type (HU).



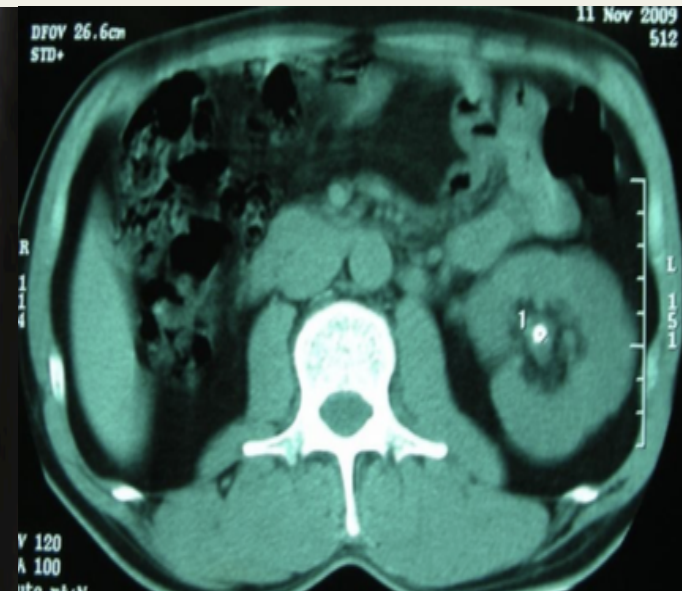




a

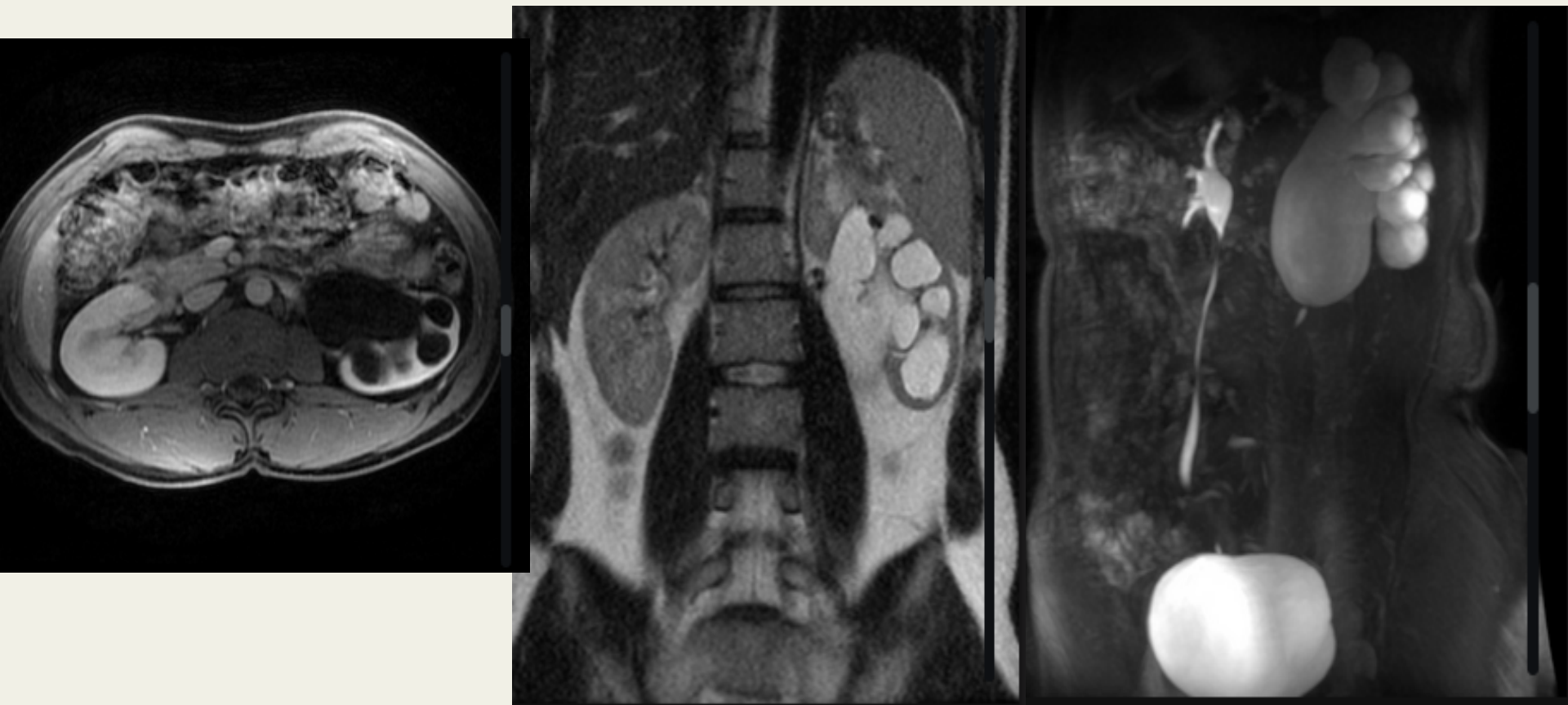


b



MRI Urography

- Magnetic resonance urography (MR urography) is a MRI study that predominantly used to image congenital abnormalities of the urinary system.
- **Indications**
- to demonstrate the urinary system or the level of obstruction in a poorly functioning kidney or urinary system obstruction
- congenital abnormalities of the urinary system
- assessment of the renal transplant donor ⁴



Pelvicureteric Junction Obstruction

Nuclear Renal Scintigraphy

- Especially in cases with signs of chronic kidney failure (CRF) to see to what extent kidney function is impaired by looking at the renogram (using a curve)
- Accurate assessment of renal function and drainage pattern
- Relative non-invasive, IV injection
- Radiopharmaceuticals → radioactive compounds used into the body.
- Gamma cameras will detect gamma rays emitted by radioisotopes

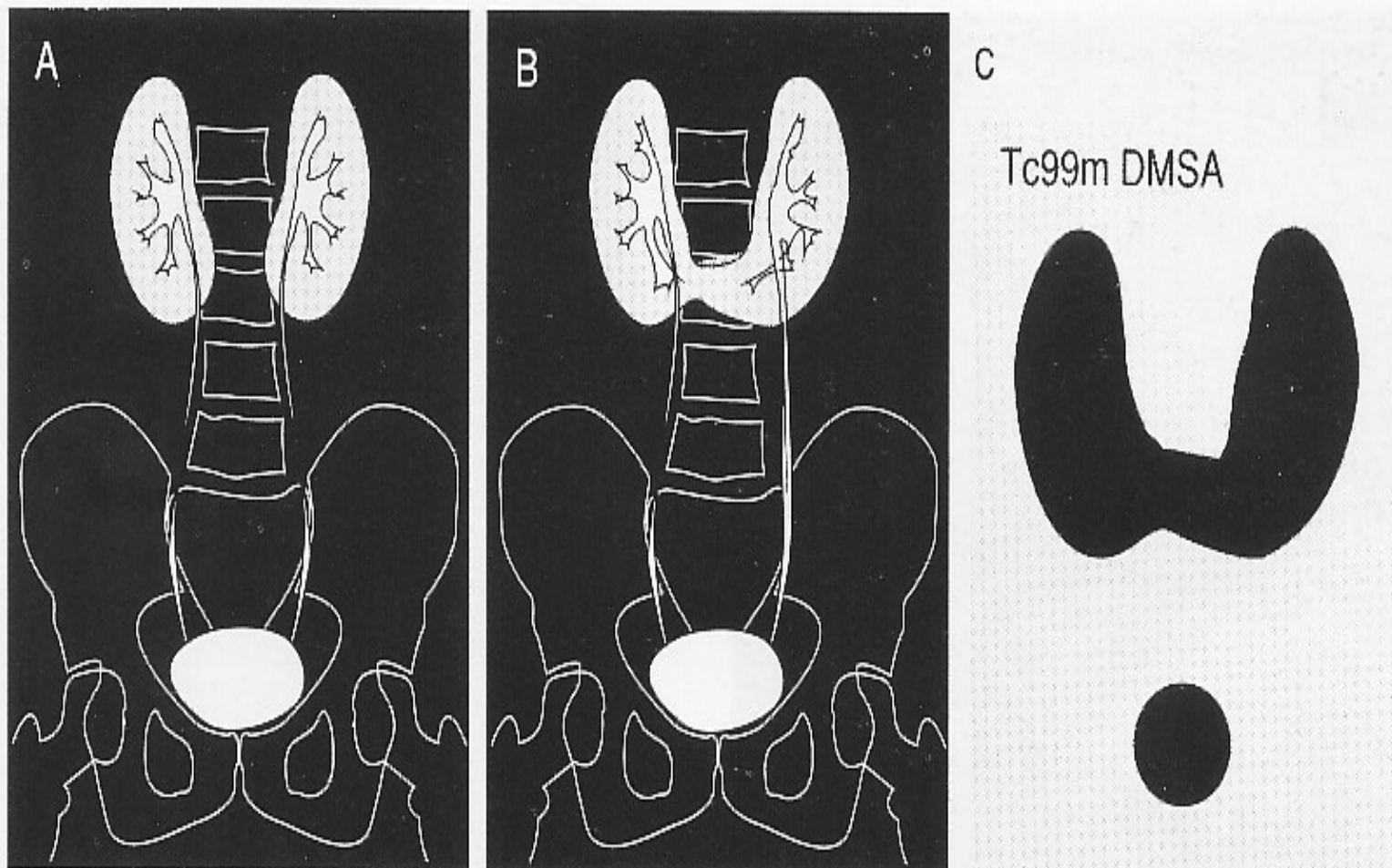
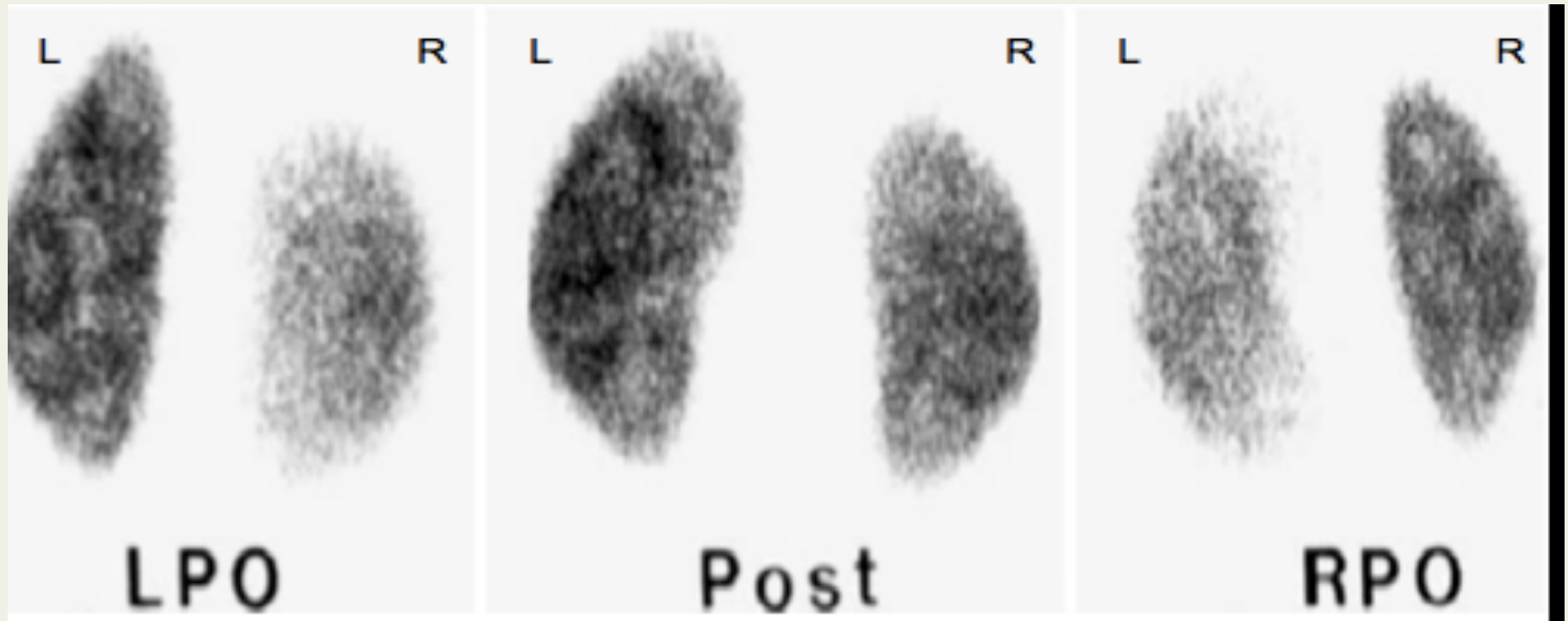


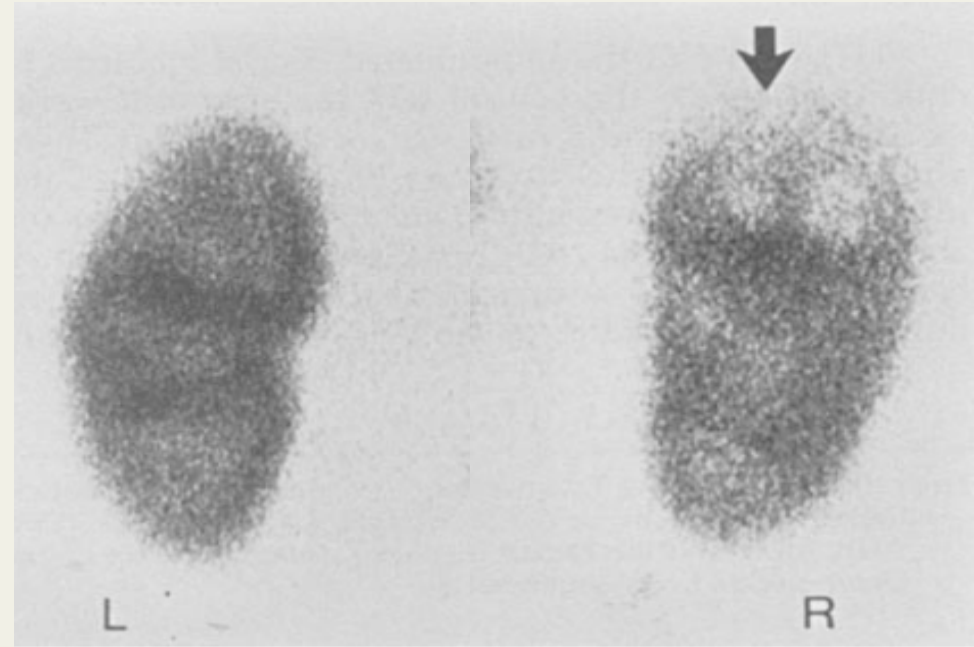
Figure 5–15. Horseshoe kidney. **(A)** Connecting isthmus is not visualized. **(B)** Isthmus is well developed. **(C)** Tc-99m DMSA scan.



Normal Tc-99m DMSA scan. This 3-year-old was undergoing investigation for a urinary tract infection. Note the clarity in the outline of both kidneys. For the right kidney, this is better on the right posterior oblique projection (*RPO*), while for the left kidney, this is best seen on the left posterior oblique projection (*LPO*) (Reproduced from *Pediatric Uroradiology*, 2nd Edition, Fötter)



Renal scar in a 3-year-old girl with previous pyelonephritis. DMSA renal scan shows focal decreased uptake with volume loss, upper pole left kidney (arrow).



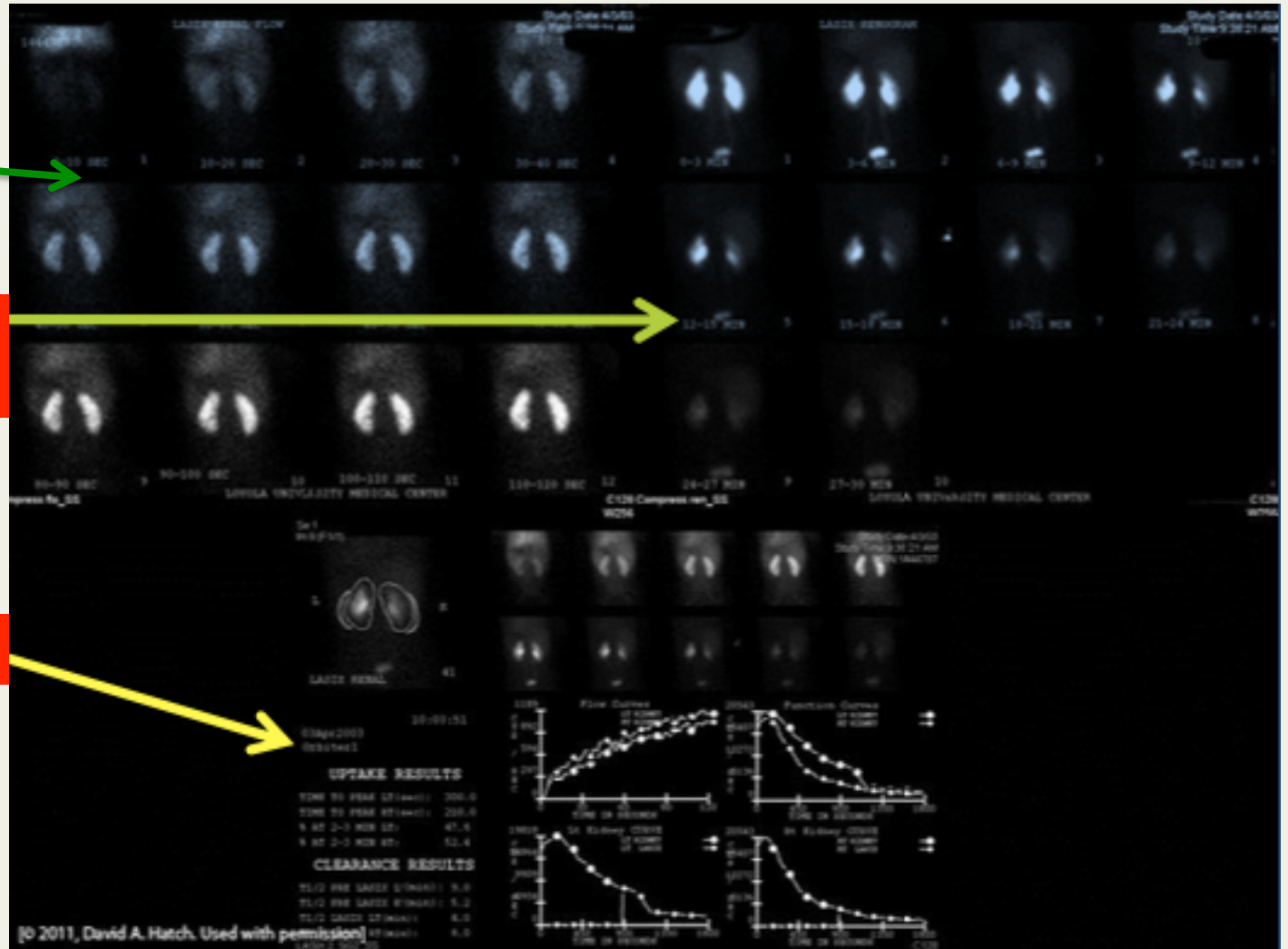
Acute pyelonephritis in a 5-month-old girl with febrile urinary tract infection. DMSA renal scan shows decreased tracer uptake without volume loss, upper pole right kidney (arrow).

Three Panels of nuclear renogram

Perfusion

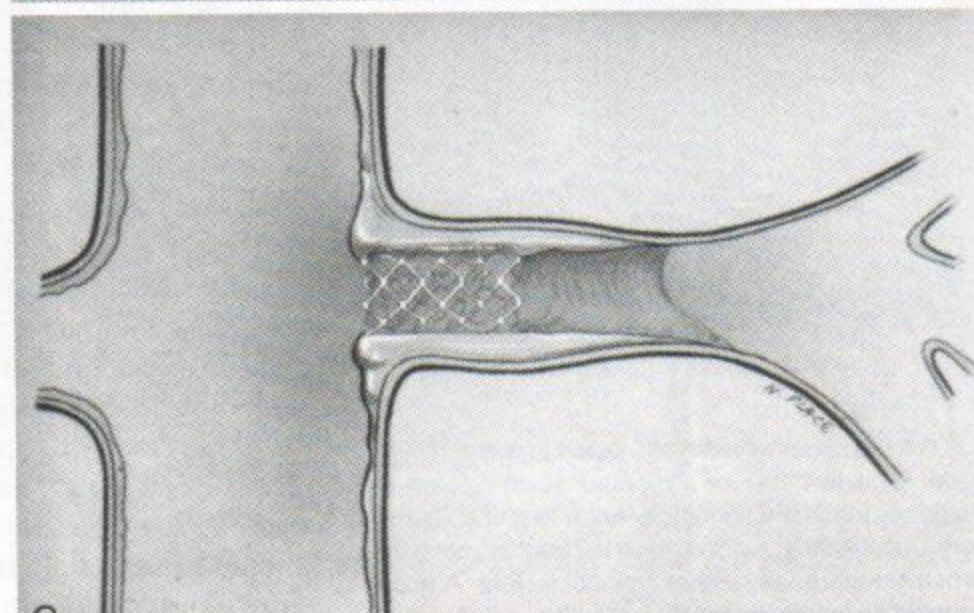
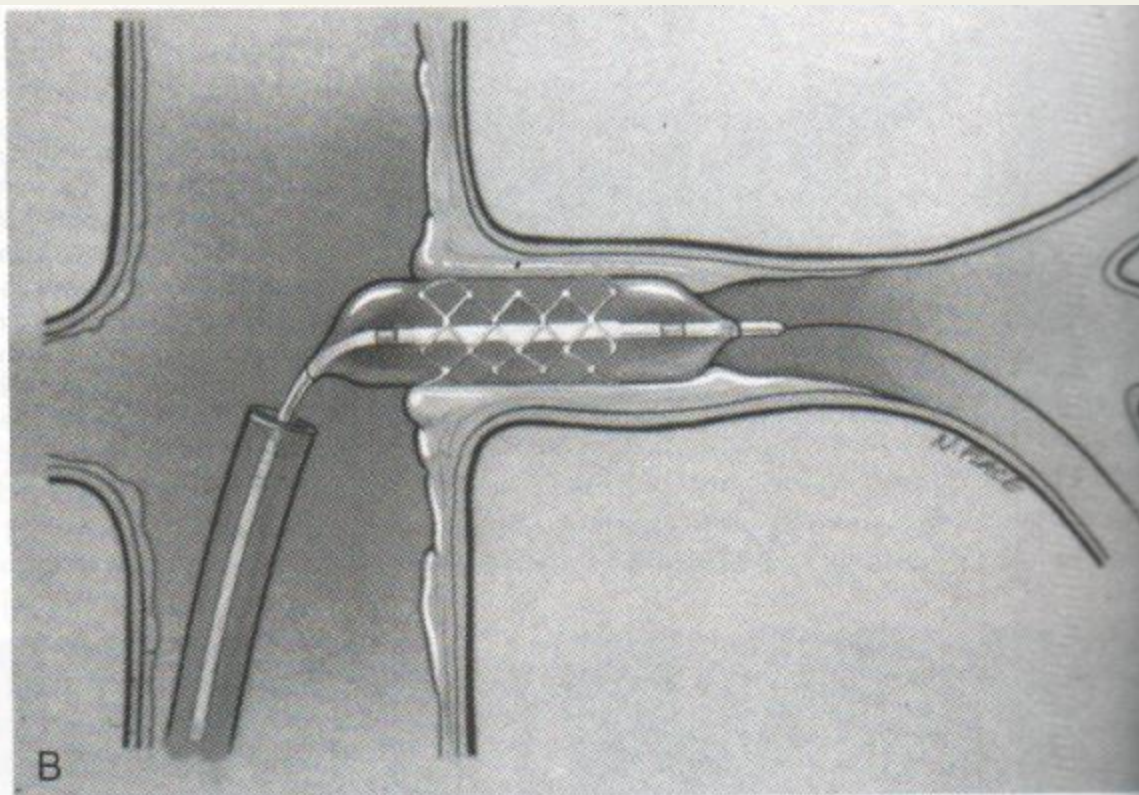
Excretion
Drainage

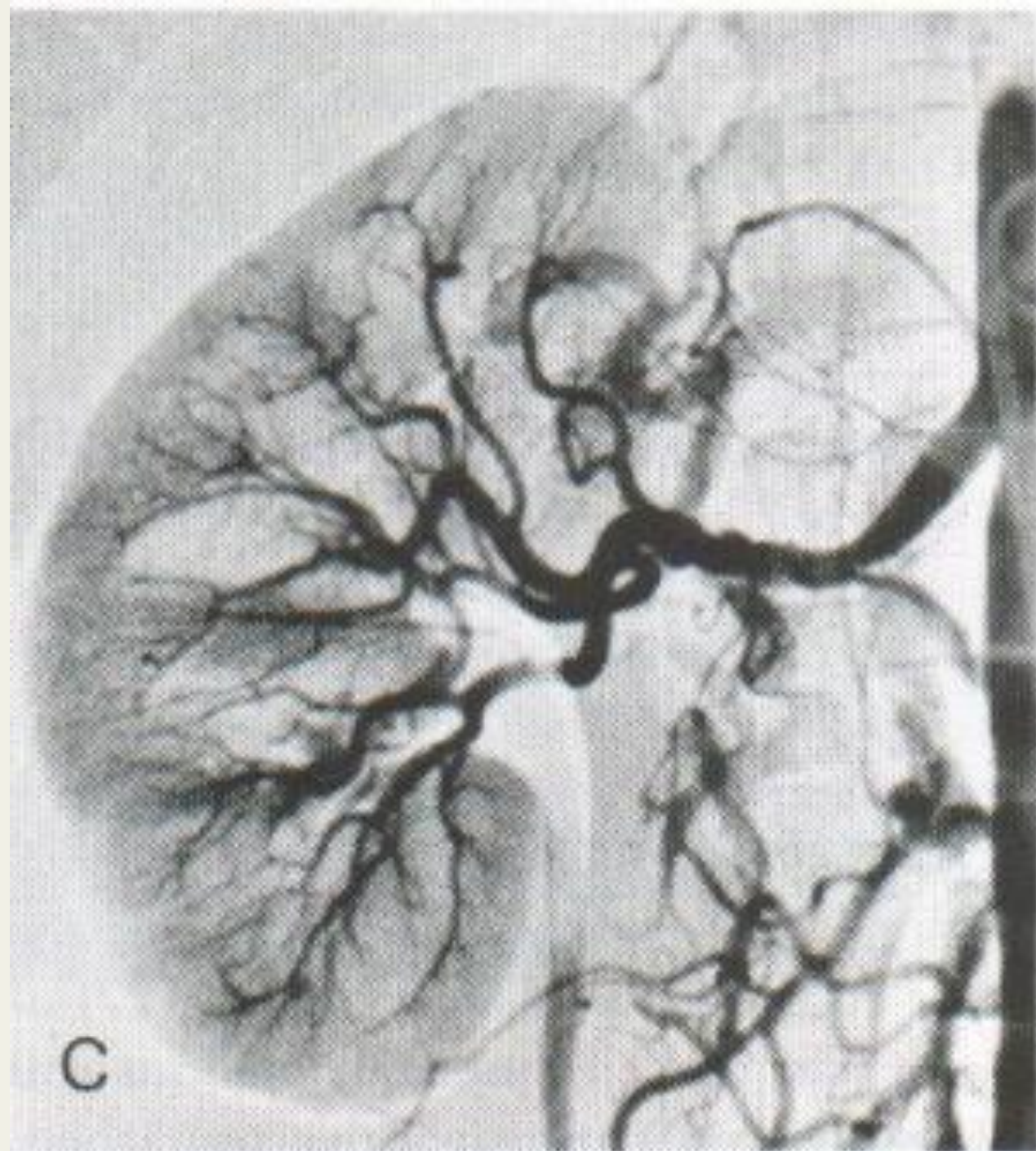
Curve



Renal arteriography

- Performed to detect vascularisation of malignant kidney tumors in preparation for surgery.
- For kidney transplantation, renal arteriography is performed for donors and iliac arteriography for recipients.
- To detect renal artery stenosis in cases of suspected renovascular hypertension, interventional radiology procedures such as ballooning and/or stenting can then be performed.







A



B

Figure 9-44. (A) Selective renal arteriogram prior to renal artery embolization. (B) Selective renal arteriogram after embolization with wire coils shows complete obstruction.

Thank you